Bright Coffee Shop Sales Analysis

Project Description

1. Introduction

The Bright Coffee Shop Sales Analysis project was undertaken to understand how a coffee shop performs in terms of sales and revenue. The project focuses on analysing sales data from January to June 2023 to identify which products sell the most, which store locations perform best, and when customers are most likely to make purchases.

As a junior data analyst, my goal was to take the available sales data and turn it into useful information that management could understand and use when making business decisions. Rather than looking at numbers alone, I wanted to understand the story behind the sales and identify areas where the business could improve.

2. Purpose of the Project

The main purpose of this project was to help Bright Coffee Shop understand its sales performance and identify opportunities to improve its operations.

The analysis aimed to answer important business questions, such as:

- Which months generated the highest and lowest revenue?
- Which product categories and individual products contributed the most to revenue?
- Which store location performed best?
- Which days of the week generated the most sales?
- What time of day was busiest?
- Do the products customers buy most frequently also generate the most revenue?
- How can the business use these findings to improve stock management, staffing, and promotions?

Answering these questions helps the business make decisions based on actual sales patterns rather than assumptions.

3. Data and Tools Used

The project used historical coffee shop sales data covering the period from January to June 2023. The dataset included information such as transaction dates, transaction IDs, quantities sold, unit prices, revenue, product categories, product types, store locations, and time classifications.

Several tools were used throughout the project, each serving a different purpose:

- **Miro:** Used to plan the project and organise the data analysis workflow and architecture.
- **Microsoft Excel:** i used excel to explore and summarise the dataset, i created PivotTables and PivotCharts, and created a sales performance dashboard.
- **SQL and Databricks:** Used to query the data, calculate sales measures, and analyse patterns across different categories.
- **Power BI:** i used PBI to create an interactive visualisation and display key performance indicators.
- **Looker Studio:** I used data studio to develop a dashboard that has sales charts, summary figures as well as filters.
- **Lovable:** i used to build a dashboard that presents sales performance and business insights visually.
- **Microsoft PowerPoint:** used to communicate my findings and recommendations in a presentation suitable for management and the ceo.

Using these tools helped me explore the data from different angles and practise presenting analytical findings in a clear and understandable way.

4. Project Approach

I began by reviewing the dataset and identifying the fields needed to answer the business questions. I then analysed revenue, quantity sold, and transaction activity to understand the overall performance of the coffee shop.

Next, I compared sales across different months, product categories, individual products, and store locations. I also examined sales by day of the week and time of day to identify periods of stronger and weaker performance.

After analysing the data, I created charts and dashboards to make the results easier to understand. The dashboards included key performance indicators and visualisations that allowed sales performance to be explored from different perspectives.

Finally, I interpreted the findings and developed recommendations that could help Bright Coffee Shop improve its stock planning, staffing decisions, promotional activities, and ongoing performance monitoring.

5. Key Findings

The analysis highlighted several important patterns in the coffee shop's sales data.

Monthly revenue: June recorded the highest revenue, while February recorded the lowest. This difference suggests that sales performance varied across the six-month period and that the business could investigate the reasons behind these changes.

Product category: Coffee was the leading product category, generating approximately 269,952, which represented 38.6% of total revenue. This shows the importance of the category to the business's overall sales performance.

Store performance: Hell's Kitchen generated the highest revenue among the store locations, with approximately 236,511. Comparing the stores can help management identify successful practices and areas where additional support may be needed.

Top-performing products: Barista Espresso was the leading product by revenue. Brewed Chai Tea and Hot Chocolate were also among the products making a strong contribution to revenue.

Sales by day: Monday recorded the highest daily revenue, while Saturday recorded the lowest in the analysis.

Sales by time of day: Morning was the strongest sales period, generating approximately 388,289, or 55.6% of total revenue. This indicates that morning demand is an important consideration when planning staffing and product preparation.

Quantity sold versus revenue: The analysis also showed that the products sold in the highest quantities do not necessarily generate the most revenue. This is important because management needs to consider both the number of units sold and the revenue earned when making decisions about product performance.

6. Business Recommendations

Based on the findings, several recommendations can be made to help Bright Coffee Shop improve its operations.

Improve stock management: The business should monitor the availability of popular products and use both quantity sold and revenue to guide stock replenishment. This can help reduce the risk of running out of products customers regularly purchase.

Plan staffing around busy periods: Since mornings generated the most revenue, management could review staff schedules and preparation activities to ensure that enough employees are available during peak demand.

Investigate store performance: Management could examine the practices at Hell's Kitchen to understand what contributes to its stronger revenue performance. Any successful practices could then be assessed for use at other locations.

Use targeted promotions: The business could test promotions during weaker sales periods to encourage purchases. The results should be measured to determine whether the promotions increase revenue without unnecessarily reducing profit margins.

Monitor sales regularly: Always updating the dashboards and reviewing the key performance indicators regularly would help management track changes and see if the sales are increasing or decreasing, identify emerging trends, and evaluate whether business decisions are producing positive results.

These recommendations should be tested and monitored over time to determine their actual impact on the business.

7. Challenges and Learning Outcomes

This project provided an opportunity to work through the different stages of a data analysis project, from understanding the dataset and writing SQL queries in databricks to building dashboards and communicating key findings.

Working with multiple tools also helped me understand that data analysis involves more than creating charts. It requires checking the data(ensuring that the data is not scrambled and its readable), calculating measures correctly, interpreting the results correctly, and explaining why the findings matter to the business and what the business can do with the findings.

The project strengthened my practical skills in data analysis, visualisation, dashboard development, and presenting insights. It also showed me the importance of checking calculations and ensuring that the information presented in a dashboard agrees with the underlying data.

8. Conclusion

The Bright Coffee Shop Sales Analysis project helped me understand how sales data can be used to evaluate business performance and support better decision-making.

The findings highlighted important patterns in monthly revenue, product performance, store results, and customer purchasing times. They also showed that looking at both quantity sold and revenue provides a more complete understanding of product performance.

Based on the analysis, Bright Coffee Shop could benefit from improving stock planning, aligning staffing with peak demand, learning from its strongest-performing store, and testing targeted promotions during quieter periods.

Overall, the project demonstrates the value of turning raw data into meaningful business insights. With regular monitoring and further investigation, these insights can help management make more informed decisions and identify opportunities for future growth.